

JAKHONGIR NODIROV

AI / Computer Vision Engineer

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SUMMARY

AI/Computer Vision engineer with 4+ years across industrial and academic settings, specializing in object detection, segmentation, and tracking, with full-cycle experience from model research through edge deployment. Deployed CNN- and transformer-based defect-inspection models on AOI equipment running 24/7 for tier-1 electronics and semiconductor manufacturers, achieving >98% defect detection at <0.5% false-positive rates. Author of two peer-reviewed publications on 3D medical image segmentation. Proficient in PyTorch, TensorRT/ONNX edge optimization, and ROS2 robotics.

TECHNICAL SKILLS

- **Languages:** Python, C/C++
- **ML/DL:** PyTorch, PyTorch3D, TensorFlow, Scikit-Learn, NumPy, OpenCV
- **Model areas:** Object Detection/Segmentation/Tracking, Medical Imaging, Anomaly Detection, Landmark & Pose Estimation, Generative/Diffusion Models
- **Deployment/MLOps:** ONNX, TensorRT, Docker/Compose, Flask, FastAPI, Gradio, Streamlit, CI/CD, AWS EC2/S3, Oracle Cloud
- **Data/Infra:** MySQL, SQLite, Weaviate
- **Edge & Robotics:** Jetson Nano, Raspberry Pi, ESP32, ROS2, MicroROS, SLAM, Nav2, Yocto, PlatformIO

WORK EXPERIENCE

HyVision System Inc. Seongnam, South Korea

AI Research Software Engineer Jun 2025 – Present

- Develop and deploy deep-learning defect-inspection pipelines for AOI (Automated Optical Inspection) equipment used in semiconductor and consumer electronics manufacturing.
- Own dataset analysis and management, model research/training, optimization, and on-device deployment across the full pipeline lifecycle.
- Document and report model performance to support production readiness and continuous improvement.

AIVAR Seoul, South Korea

AI Research Software Engineer Feb 2023 – Mar 2025

- Led development of a 3D human avatar reconstruction server application using 3D morphable face and body models, deployed on AWS with Docker.
- Built and shipped a deep-learning-based image verification system (vector embedding search) for a government-backed digital IP-protection project.
- Researched and optimized CNN- and transformer-based vision models; managed deployment and monitoring of AI applications in production.

Gachon University — CT Lab Seongnam, South Korea

Graduate Researcher Sep 2020 – Feb 2023

- Led research and development of attention-based UNet and 3D-UNet CNN architectures for volumetric medical image segmentation, published in two peer-reviewed venues.
- Managed and contributed to applied CV lab projects including road-condition assessment and license-plate recognition.

SELECTED PROJECTS

Semiconductor Wafer Defect Inspection *HyVision, 2025*

- Built a high-precision defect inspection pipeline for 8-inch and 12-inch silicon wafers using CNN-based object detection, segmentation, and anomaly detection models, deployed on industrial edge GPUs inside wafer inspection equipment.

- PyTorch, ONNX/TensorRT edge optimization · Runs 24/7 in production: >98% detection rate on critical defects at <0.5% false-positive rate.

Consumer Electronics Surface Inspection *HyVision, 2025*

- Built an end-to-end automated defect inspection pipeline (segmentation, detection, classification) for glass and metal enclosure parts, deployed on AOI machines in a tier-1 consumer electronics client's production line.
- PyTorch, ONNX/TensorRT edge optimization · Sub-millimeter defect detection at line speed with a low false-positive rate, directly contributing to yield improvement.

Driver Monitoring System (DMS) *Lab Project*

- Upgraded grayscale head-pose, facial landmark, and seatbelt-detection models using CNN- and keypoint-based architectures; retrained on an augmented grayscale dataset and deployed via ONNX/TensorRT on NVIDIA Jetson for day-and-night reliability on edge hardware.

3D Human Avatar Reconstruction *AIVAR, 2023–2024*

- Converted 2D images into textured 3D face/body avatars using 3D morphable face and body model architectures, with diffusion-based image correction and neural texture-matching; deployed on AWS with Docker.

IP Protection — Image Verification *AIVAR, 2024*

- Built a vector-embedding image verification system for animation-character and logo IP protection using CNN-based face/feature recognition models, with a demo web application; part of a government-backed digital IP-protection project.

Attention 3D U-Net for Brain Tumor Segmentation *CT Lab, 2021–2022*

- Designed a novel attention-based 3D U-Net with multiple skip connections for volumetric MRI tumor segmentation, improving boundary-detection precision. Published in *Sensors* (2022).

Autonomous Driving Robot (ROS2) *Side Project*

- Built a mobile robot on Jetson Nano B1 with ROS2 Foxy, motor control, encoder feedback, RP-Lidar, and a web GUI (roslib.js) for remote operation; advanced SLAM/navigation capability.

EDUCATION

Gachon University 2020.09 – 2022.08

Master's Degree, Computer Engineering — GPA 4.17/4.5

PUBLICATIONS

- "Attention 3D U-Net with Multiple Skip Connections for Segmentation of Brain Tumor Images." *Sensors* 22, no. 17 (2022): 6501.
- "3D Volume Reconstruction from MRI Slices based on VTK." In *2021 International Conference on Information and Communication Technology Convergence (ICTC)*, pp. 689–692. IEEE, 2021.

LANGUAGES

Uzbek (Native) · English (Proficient) · Korean (Intermediate)